Chapter 16 worked solutions • Virtual machines

OS16-E01 • Modes and virtual interrupts

1. No. VMM updates guest virtualIF and interrupt injection while host serves others. A guest application lacking virtual kernel privilege receives a guest exception; trapping does not grant rights. Ring0 is not root context.
2. VMM cannot update state, allowing observable semantic error. Translation, cooperative hypercalls, or hardware interception can mediate. Absence of trap does not prove absence of required behavior.

OS16-E02 • Addresses and permission intersection

1. Displacement0xABC gives GPA0x7ABC, HPA0x12ABC. Read passes both; EPT denies write despite guest permission.
2. Access can reach the old host page. Track guest updates, rebuild the shadow composition, and invalidate old translations under architecture rules. Hardware two-stage also needs its own cache/invalidation rules. GVA, GPA, HPA are distinct.

OS16-E03 • General nested-walk cost

1. Each guest PTE costs 4 EPT+1 guest read; three cost15. Final GPA adds4, total19 table references plus one data=20. Formula$(m+1)(n+1)-1$.
2. Every guest PTE itself resides in guest physical memory and needs second-stage translation. Caches lower the bound;19 is not every load's fixed cost. Separate table and data references.

OS16-E04 • Four-slot PV ring

1. Queued$7-6=1$, outstanding$6-5=1$, response$5-4=1$, used$7-4=3$, free1. Next slot$7 \bmod 4=3$. Use monotonic counts, not differences of modulo pointers.
2. Submission makes req_prod8, used4, full. Consuming response advances rsp_cons$4\to5$, used3, free1, allowing one more. Advancing req_cons alone releases no frontend capacity.
3. Fill descriptor then release-publish req_prod; backend acquire-observes before reading. Fill response before publishing rsp_prod; frontend consumes then advances rsp_cons. Notifications replace neither predicates nor ordering. Buffers live through completion. Out-of-order completion needs another model.

Chapter 16 worked solutions • Virtual machines (continued)

OS16-E05 • Finite two-layer scheduling trace

1. A runs0–10 and20–30, finishing30. B runs10–20 and30–40, finishing40. Equal long-run50% shares have different phases and do not imply40 ms for every20 ms budget.
2. No. Guest schedules within vCPU allocations; it cannot force host pCPU service. Stronger latency requires host real-time/isolation/admission guarantees. CPU runtime and wall response differ.

OS16-E06 • Balloon conservation

1. Demand8192 exceeds guest-available7168 by1024. Ballooning512 each gives targets3584+3584=7168, plus VMM 1024=8192. This conserves targets, not a claim that original guest 8192 plus VMM 1024 already fit. Inflation reduces backing demand without creating RAM.
2. No. Driver reserves guest pages and reports them; VMM reclaims backing. Unexpected access must fault/reback rather than reach another guest's frame. Cache reclamation/pageout costs remain. Guest reservation does not mean host backing is unreclaimable.

OS16-E07 • Sharing and COW

1. Hash only selects candidates. Compare full bytes with safe synchronization, then map both read-only to h9. On B's fault, allocate h12, copy X, change mapping/permissions and invalidate as required, then permit Y. A retains X. Never write first and copy later.
2. The old translation can still modify shared data, breaking A's isolation. Required TLB/second-stage invalidation and synchronization precede safe sharing. Contents, mappings, references, and permissions must agree.

Chapter 16 worked solutions • Virtual machines (continued)

OS16-E08 • Migration convergence and pause

1. R0=512 copies4 s→R1=128; next1 s→R2=32; next0.25 s→R3=8. Freeze copies8 in 0.0625 s. Total680 MiB,5.3125 s, pause0.0625 s.
2. Ratio=d/B=1, so512 does not shrink. Bound rounds, reduce dirtying, extend pause, or change strategy. Short downtime is not universal.
3. Two active copies with one identity can create duplicate effects, or split brain. Explicit ownership/commit/recovery is required. A normal trace does not prove partition safety. Compatible CPU/devices and shared storage are this model's premises.

OS16-E09 • Snapshots and external effects

1. Active read 0=A, read 1=C; base restores read 1=B. Mail remains sent, and rerunning can send twice. A snapshot is not an external-world rollback.
2. No. Files/chains/RAM/disk can represent different times. Shutdown or supported consistent snapshot plus application quiescence and all required metadata/chains is needed. Same-failure-domain snapshots are not independent backups. Crash and application consistency differ.

OS16-E10 • Abstractions and bytecode

1. Another kernel requires system VM/emulator; cross-ISA binary needs emulation/translation; class requires JVM/runtime; same-kernel separation can use containment. Containers share kernel, system VMs have guest kernels, and JVM is not another general OS.
2. []→[2]→[2,3]→[5]→[], local1=5. Iadd needs two integers; empty stack violates verifier premises rather than authorizing arbitrary memory access. Type checking does not prove business authorization. The interface/isolation layers determine guarantees.

Chapter 16 worked solutions • Virtual machines (continued)

Verification code appendix: vm_models.py

```
from fractions import Fraction as F
def translation(va,gpt,ept):
 vpn,offset=divmod(va,4096);gfn,gperm=gpt[vpn];mfn,hperm=ept[gfn]
 return (gfn*4096+offset,mfn*4096+offset,gperm&hperm)
def ring(prod,cons,rprod,rcons,n):
 assert rcons<=rprod<=cons<=prod and 0<=prod-rcons<=n
 return prod-cons,cons-rprod,rprod-rcons,prod-rcons,n-(prod-rcons)
def precopy(memory,bandwidth,dirty,rounds):
 r=F(memory);trace=[]
 for i in range(rounds):
  t=r/bandwidth;trace.append((r,t));r=dirty*t
 return trace,r,r/bandwidth
if __name__=='__main__':
 assert translation(0x1234,{1:(5,{'r','w'})},{5:(10,{'r'})})==(0x5234,0xA234,{'r'})
 assert (4+1)*(4+1)-1==24 and (3+1)*(4+1)-1==19
 assert ring(7,6,5,4,4)==(1,1,1,3,1)
 assert ring(8,6,5,4,4)[-1]==0 and ring(8,6,5,5,4)[-1]==1
 t,r,p=precopy(1024,256,64,3);assert t==[(1024,F(4)),(256,F(1)),(64,F(1,4))]
 assert r==16 and p==F(1,16) and sum(x[0] for x in t)+r==1360
 t2,r2,p2=precopy(512,128,32,3);assert r2==8 and p2==F(1,16)
 assert precopy(512,128,128,3)[1]==512
 assert 2304-256+2304-768+512==4096
 print('translation',hex(0x5234),hex(0xA234),'permissions r only')
 print('pre-copy rounds',t,'remaining',r,'pause',p,'s; total bytes 1360 MiB')
```

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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